

War room reference

Team PR - Saturday 26 September 2026 - mentors. Say "our app", never the name. "Where" = nationwide, all nine provinces. Always name spaza shops AND builders.

1. The 45-second intro (say it first)

"We're building a mobile app for small businesses and informal traders - spaza shops, builders and other informal businesses - to run their business and order from the suppliers they already work with. Everyday business tools - a credit book, an order book, jobs for builders - help them run the business, and they tell us, privately, what the business needs, so we can recommend the right suppliers. The businesses use it free; suppliers are our paying customers, because most small businesses already have suppliers they trust. We're not a marketplace: we recommend suppliers, never products. It's built and tested end to end: sign-up, suppliers recommended with reasons, connect, order, pay digitally or cash, invoices and receipts with a QR check - and the tools."

Honest wording (checked in the code today): the tools already produce private demand signals (e.g. "open jobs need plumbing"); feeding them into the supplier ranking is the next piece we're building tonight. If asked "does it use the tools yet?" say exactly that. Don't say "proof" about tool records.

2. Our 3 asks (the point of the session)

Ask	What to say	Why it matters
1 Business model	"Suppliers pay us. Which works best to start: a monthly subscription, a small fee per order, or clearly labelled paid placement? What would a township supplier actually pay?"	Business + pitch: 15 points
2 Pitch + demo	"We have more than 3 minutes of features. Which ONE flow should the 90-second video show: a supplier order paid digitally with a verified invoice, or a spaza / builder using a tool that shapes the supplier recommendation?"	Sun 10:00 pitch; top 12 demo
3 TRL	"We think we're TRL 4-5: validated end to end in our own environment, payments in the provider's sandbox, no real users yet. Is that how you'd rate it, and what gets us to 6?"	Final submission needs a TRL statement

Spare asks

- "Would suppliers trust an engine that ranks them? We rank every supplier, never hide one, and show the reasons. Is that enough?"
- "What makes a supplier sign up first - the order flow, or seeing demand from the tools?"
- "Lean Canvas (due 21:30): does our unfair advantage hold up?"
- "What proof of demand convinces judges in a weekend: interviews, a supplier letter of intent, a pilot sign-up?"

3. What we're building

The problem. Spaza shops, builders and other informal businesses run on calls, WhatsApp, cash and a notebook. Nothing leaves a record both sides accept: suppliers can't reconcile cash, builders and clients argue over what was paid, credit to customers lives in memory, and the owner can't show the business is real.

The product. One free app to run the business and order from the suppliers you already use. Hybrid: the businesses are the users (free); suppliers are the paying customers. Not a marketplace.

How it fits together

Step	What happens
1 Sign up	Email + password, 6-digit code by email; business name, type (spaza, builder + trade, food...), where you are (map pin), what you buy and how you pay. Every step saves as you go.
2 Suppliers	Recommendation engine ranks EVERY supplier with plain reasons: what you buy (40), reach - delivers to you 35 / you can collect 25 (35), closeness (10), how you pay (10), minimum order fits your spend (5). Honest reasons too ("outside their delivery area").
3 Connect + order	Connect to a supplier, browse its catalogue (server prices, stock held atomically), cart, checkout: delivery to the business or a saved address, or collect.
4 Pay	Digital via PayFast (sandbox) - confirmed only after 4 checks (signature, PayFast IP, amount, validate back) - or cash on delivery, capped at R1,000 per order and 2 waiting.
5 Documents	Invoice (tax invoice when the supplier is VAT-registered) and receipt, each with a QR code to a public check page that shows no personal details.
6 Tools	Credit book, order book (counter orders, works offline), jobs (stages, camera-only photos, client sign-off by a one-use link), builder network (find and bring partners). Each tool produces private demand signals for the engine.

Who it's for

- **Spaza shops:** credit book (who owes me, repayments), order book for counter orders, restock from wholesalers.
- **Builders and trades** (11 trades: bricklayer, plumber, electrician, roofer, tiler, painter, welder, glazier...): jobs in stages with client sign-off, materials from hardware suppliers, a partner network.
- **Food sellers and other informal businesses:** order book + suppliers.
- **Suppliers** (wholesalers, cash-and-carries, hardware): more orders from the businesses they already serve, fewer cash disputes, a view of demand. They connect through an integration feed (checked row by row); 11 sample suppliers, 1,672 products.

4. Built and tested vs. in progress (say it honestly)

Area	Built and tested	In progress / next
Account	Email-verified sign-up, code on a new phone, trusted phone, sessions, sign out other phones, data export, account closure, consent recorded	Google sign-in (backend done; app needs a new build)
Business profile	Onboarding saves step by step; edits save with a clear 'Saved' or the server's reason; tools on/off	-
Suppliers	Feed, engine with reasons, connect, catalogue, cart, map address search	Tools' demand signals in the ranking
Orders	Server pricing, atomic stock, retries never order twice, cash caps, saved delivery addresses (max 5, default)	"I paid the cash"; supplier-side app
Payments	PayFast sandbox, 4-check notification, paid order confirmed at once, payment email	Split payouts to suppliers
Documents	Invoices, receipts, QR check page (no personal data)	-
Tools	Credit book (+ history, bin), order book (offline outbox), jobs (photos, sign-off), builder network, demand signals	Daily tally
Home / Account	Home opens tools; live credit / jobs / order-book numbers	Hide balances, real 'you owe suppliers', My record screen
Notifications	Stored alerts in two tabs (Orders, Inbox), read state, switches, rate limits	Alerts from orders/sign-ins; app screen; phone push later

Numbers to quote

- **1,178 automated backend tests + 155 app tests**, all passing; TypeScript and lint clean; run before every merge.
- Hostile-input tests on every endpoint (bad ids, unknown fields, huge / nested JSON, NUL characters, someone else's data).
- 24 database migrations; 11 sample suppliers with 1,672 products; 11 trades.

Honest gaps (we list them, we don't hide them)

- Cloudflare / WAF in front (plan written; ngrok for the demo), shared rate-limit store (Redis), malware scanning of uploads switched on.
- Two-phone cash handshake ("confirmed by both"), split payouts to suppliers, supplier API keys for ERP.
- Offline encrypted store on the phone, phone push notifications, external penetration test, POPIA legal review.

5. How it's built + security

Layer	What we use
App	Expo SDK 57 / React Native (Android dev build), expo-router typed routes, TypeScript; sample-data mode for demos
Backend	Python 3.12, Flask 3.1, SQLAlchemy 2, Alembic migrations, marshmallow strict schemas
Data	PostgreSQL 16+, separate schemas: identity, trader, supplier, commerce, platform, audit
Services	PayFast (payments), Cloudinary (signed direct uploads), Mapbox via our backend (geocoding), email
Structure	Domains -> features (api, models, repositories, schemas, services). Shared code never imports a domain; no feature reads another's tables; identity is sealed. An architecture test enforces it.

Security by design (SSDLC submitted)

- **IDOR**: every query scoped to the signed-in user; someone else's id answers "not found" (404), tested per feature.
- **Input**: strict schemas refuse unknown fields; cleaned text; money in integer cents; pins inside South Africa.
- **The server prices everything**: the app sends products and quantities only.
- **Rate limits** per user / IP on every route; **idempotency keys** so a retry on bad signal never doubles an order or sale.
- **Audit trail** of disputable moments with ids and amounts only - no names, phones or addresses. Client IP can't be forged.
- **Payments**: card data never touches us (PayFast hosted page); notifications trusted only after 4 checks.
- **Honest proof**: only a provider-verified digital payment is proof; cash is at most "confirmed by both"; tool records are the owner's own records; never added together.
- **Privacy**: the engine uses aggregates only, private to the business; rule-based, not trained on people's data (no membership-inference risk). Data export and account closure built in. OWASP API Top 10 mapped; STRIDE threat model; incident plan.

6. Likely questions - short answers

They ask	We answer
Who pays?	Suppliers. The businesses use it free.
Why suppliers?	Most small businesses already buy from suppliers they trust; we make that relationship work better - more orders, fewer cash disputes, a view of demand.
Isn't it a marketplace?	No. Businesses order from suppliers they connect to. We recommend suppliers, never products.
How is this better than WhatsApp orders?	Prices from the supplier's own catalogue, stock checked, payment verified, a proper invoice and receipt, and a record of what you owe and are owed.
What stops fake orders?	Only connected suppliers; cash capped at R1,000 and 2 waiting; digital is paid up front.
Is the tool data proof?	No - it's the business's own record. Only a digital payment confirmed by the provider is proof; every order says which it is.
How does the engine use their data?	Aggregates only ("open jobs need plumbing"), private to that business, never shared with suppliers. Rule-based scoring with reasons you can read.
Competitors?	Each does one slice: Spazapp (bulk buying), Kasi App (POS), iKhokha (card payments), Kazang / Flash (airtime & VAS float), Kasi Money (banking), HustleCheck (reviews). None combines the everyday tools with ordering from the suppliers you already use. We complement Kazang / Flash rather than replace them.
Will traders switch from WhatsApp and the notebook?	Free tools that help every day, with the same suppliers - just easier. Works on a cheap Android; the order book works offline.
Commission on thin margins?	About 1% only on digital orders; cash stays free.
Security?	SSDLC submitted: threat model, OWASP API Top 10, data inventory, card data never touches us, IDOR tests on every feature, incident response plan.
What's next after the hackathon?	Pilot with 1-2 suppliers and the businesses they already serve; Cloudflare, split payouts, offline tools, POPIA review.

7. The market and the money

The problem in numbers

- Township economy about **R900 billion a year**, **R180 billion** through spaza shops (Standard Bank, 2025).
- About **80%** of township businesses unregistered; **fewer than 9%** can get a bank loan; **57%** run on personal savings (Standard Bank, 2025).
- Spaza Shop Support Fund: R500 million opened, only **R179.6 million** reached shops by May 2026; **354** applications failed verification (DSBD, 2026).
- About a third of South Africa's jobs are informal; construction employs about 1.2 million, 38% informally (ILO, 2024).

Business model (Lean Canvas)

Stream	Price
Businesses	Free, always, for the core tools
Supplier Starter	R299 / month
Supplier Pro	R1,499 / month
ERP integration (later)	R4,999 / month + R15,000 setup
Digital orders	About 1% (none on cash)

- **Running costs:** pilot about R460 / month; growth about R4,000-R4,700 / month; Google Play R415 once.
- **Pilot ask:** R300,000 for 6 months (field agent, stipends, data bundles, running costs, POPIA review + external security test, devices).
- **Break-even (est.):** 2 suppliers on Starter in the pilot; about 4 on Pro at growth.
- **Pilot metrics (est.):** 2 suppliers + 50-100 of their businesses; 60% weekly active; 30% of their orders in the app; 50% paid digitally.
- **Distribution:** supplier-led (each supplier onboards the businesses it serves), business-led invites, builders bring partners.

8. Rules for the call

- Don't say the product name on shared calls or in chat until Sunday's submission - "our app".
- "Where": nationwide (all nine provinces), never our test places.
- Spaza shops AND builders in every example; never "street vendors".
- Honest about built vs planned. One talks, one takes notes.

9. Notes from the war room

Mentor feedback:

Answers to ask 1 (model) / 2 (demo flow) / 3 (TRL):

To do tonight (Lean Canvas 21:30, slides, video, submission Sun 09:00 - aim 08:00):